How-To Library

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Oracle Database Privilege Requirements in Test Data Management

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Abstract

You can use an Oracle database to create connections in Test Data Management (TDM). You can create connections that you use at different stages in a TDM task. The database user might require different privileges based on the type of connection. This article describes the privileges required for different types of connections and the SQL commands you can run to assign the privileges. The article assumes that you have knowledge of Oracle databases, SQL commands, TDM, and are familiar with TDM operations.

Supported Versions

- Test Data Management 10.2.0 HotFix 1
- Test Data Management 10.2.1
- Test Data Management 10.2 R1
- Test Data Management 10.2.0
- Test Data Management 10.1.0

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Overview

You can create connections using different databases and use them to perform different tasks in Test Data Manager.

For example, you can create connections to databases to perform data subset, data masking, and data generation operations, and to run profiles for data discovery operations. TDM supports some databases for certain TDM operations only. When you create a connection, relevant properties appear in the connection configuration wizard.

You can use Oracle database connections as the following connections in a plan:

- Source connection
- Target connection
- Staging connection
- Storage connection

To use a database as a connection in a plan, the database user must have the required privileges. A TDM operation might fail if the user does not have the minimum required privileges.

The minimum privileges required differ based on whether you use the database as a source, target, staging, or storage connection. You can choose to assign all required privileges to the same user or to create different users for different types of connections.

Note: The privileges mentioned in the document are based on Oracle Database 12c Release 2. If the same privileges are not available on other Oracle versions, you can consider equivalent privileges.

Single User for All Connection Types

You can choose to use a single database user to access all connection types.

A single user who has all privileges or all required privileges can perform tasks that include the following Oracle connections in a plan:

- Source connection
- Target connection
- Staging connection
- Storage connection

Run the following SQL command to assign all privileges:

```
GRANT ALL PRIVILEGES TO <USER>
```

`<USER>` is the database user name.

Source Connection

You can use an Oracle database as a source connection.

You require the following privileges to use an Oracle database as a source:

- CREATE SESSION
- SELECT on source tables

Run the following SQL commands to assign privileges:

- `GRANT CREATE SESSION to <USER>;`
- `GRANT SELECT ON <TABLE> TO <USER>;`

`<USER>` is the database user name and `<TABLE>` is the table name.

Target Connection

You can use an Oracle database as a target connection.

You require the following privileges to use an Oracle database as a target:

- CREATE SESSION
- UNLIMITED TABLESPACE
- ALTER ANY TRIGGER
- ALTER ANY TABLE
- DROP ANY INDEX
- CREATE ANY INDEX
- ALTER ANY INDEX
- EXECUTE
  EXECUTE object privilege required on user-defined functions used in the function-based index if the functions are owned by another user.
- CREATE SEQUENCE
- ALTER

ALTER privilege required on Sequence created internally by the database for Identity columns.

- `SELECT SEQUENCE_NAME FROM ALL_TAB_IDENTITY_COLS WHERE TABLE_NAME = '<TABLE>' AND OWNER ='<OWNER_NAME> ';`
  ALL_TAB_IDENTITY_COLS is a system table.
- `GRANT ALTER ON <SEQUENCE_NAME> TO <USER>;`

Note: `<TABLE>` is the table name and `<OWNER_NAME>` is the database owner name. `<SEQUENCE_NAME>` is the sequence name of each sequence that the first command returns. Run the second command for all sequences listed when you run the first command.

- SELECT
  Required on the following System tables:
  - ALL_CONSTRAINTS
  - ALL_TABLES
  - ALL_INDEXES
  - ALL_IND_COLUMNS
  - ALL_IND_EXPRESSIONS
  - ALL_PART_INDEXES
  - ALL_IND_PARTITIONS
  - ALL_PART_KEY_COLUMNS
  - ALL_TAB_IDENTITY_COLS
  - ALL_CONS_COLUMNS
  - ALL_TRIGGERS
- The following privileges are required on target tables:
  - SELECT
  - INSERT
  - UPDATE
  - DELETE

Note: On Oracle Database 12c Release 2, you require SELECT privilege to update tables.

Run the following SQL commands to assign privileges:

- `GRANT CREATE SESSION to <USER>;`
- `GRANT UNLIMITED TABLESPACE TO <USER>;`
- `GRANT ALTER ANY TRIGGER to <USER>;`
- `GRANT ALTER ANY TABLE to <USER>;`
- `GRANT DROP ANY INDEX to <USER>;`
- `GRANT CREATE ANY INDEX to <USER>;`
- `GRANT ALTER ANY INDEX to <USER>;`
- `GRANT CREATE SEQUENCE to <USER>;`
- `GRANT SELECT ON ALL_CONSTRAINTS TO <USER>;`
- `GRANT SELECT ON ALL_TABLES TO <USER>;`
- `GRANT SELECT ON ALL_INDEXES TO <USER>;`

- `GRANT SELECT ON ALL_IND_COLUMNS TO <USER>;`
- `GRANT SELECT ON ALL_IND_EXPRESSIONS TO <USER>;`
- `GRANT SELECT ON ALL_PART_INDEXES TO <USER>;`
- `GRANT SELECT ON ALL_IND_PARTITIONS TO <USER>;`
- `GRANT SELECT ON ALL_PART_KEY_COLUMNS TO <USER>;`
- `GRANT SELECT ON ALL_TAB_IDENTITY_COLS TO <USER>;`
- `GRANT SELECT ON ALL_CONS_COLUMNS TO <USER>;`
- `GRANT SELECT ON ALL_TRIGGERS TO <USER>;`
- `GRANT SELECT, INSERT, UPDATE, DELETE ON <TABLE> TO <USER>;`

`<USER>` is the database user name and `<TABLE>` is the table name.

Staging Connection

You can use an Oracle database as a staging connection.

You require the following privileges to use an Oracle database as a staging connection:

- CREATE ANY TABLE
- UPDATE ANY TABLE
- ALTER ANY TABLE
- SELECT ANY TABLE
- INSERT ANY TABLE
- DROP ANY TABLE
- DELETE ANY TABLE
- CREATE ANY INDEX
- CREATE ANY SEQUENCE

Run the following SQL commands to assign privileges:

- `GRANT CREATE ANY TABLE to <USER>;`
- `GRANT UPDATE ANY TABLE TO <USER>;`
- `GRANT SELECT ANY TABLE TO <USER>;`
- `GRANT DROP ANY TABLE TO <USER>;`
- `GRANT DELETE ANY TABLE TO <USER>;`
- `GRANT CREATE ANY INDEX TO <USER>;`
- `GRANT ALTER ANY TABLE TO <USER>;`
- `GRANT INSERT ANY TABLE TO <USER>;`
- `GRANT CREATE ANY SEQUENCE TO <USER>;`

`<USER>` is the database user name.

Storage Connection

You can use an Oracle database as a storage connection.

You require the following privileges to use an Oracle database as a storage connection:

- CREATE ANY TABLE
- SELECT on storage tables IDM_EXPRESSION_STORAGE, IDM_SUBSTITUTION_STORAGE
- INSERT on storage tables IDM_EXPRESSION_STORAGE, IDM_SUBSTITUTION_STORAGE

Run the following SQL commands to assign privileges:

- `GRANT CREATE ANY TABLE TO <USER>;`
- `GRANT SELECT, INSERT ON IDM_EXPRESSION_STORAGE TO <USER>;`
- `GRANT SELECT, INSERT ON IDM_SUBSTITUTION_STORAGE TO <USER>;`

`<USER>` is the database user name.

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